

MOBILE APPLICATION DEVELOPMENT

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Instructor Information

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Let's Start

Lecture # 3

OUTLINE

- Introduction of Material APP
- What is Widgets
- Types of Widgets
- Widget Tree



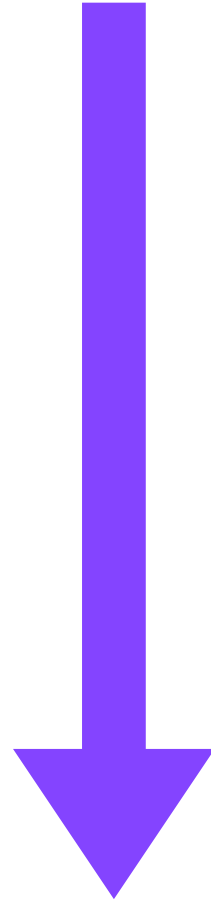
ABOUT MATERIAL DESIGN

Google's flexible design system

A set of suggestions, rules & guidelines that help you build beautiful user interfaces

Highly customizable and extendable

HOW FLUTTER APPS BECOME ACTIVE



1

`main()` function gets executed automatically

By Dart, when executing the compiled app on the target device

2

`runApp()` should be called inside of `main()`

`runApp()` "tells" Flutter what to display on the screen (i.e., which UI elements to display)

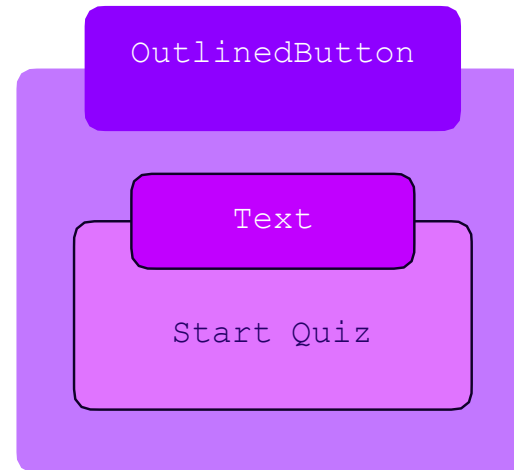
3

Pass the to-be-displayed "widget tree" to `runApp()`

A "widget tree" is a combination of (nested) Flutter widgets that build the overall user interface

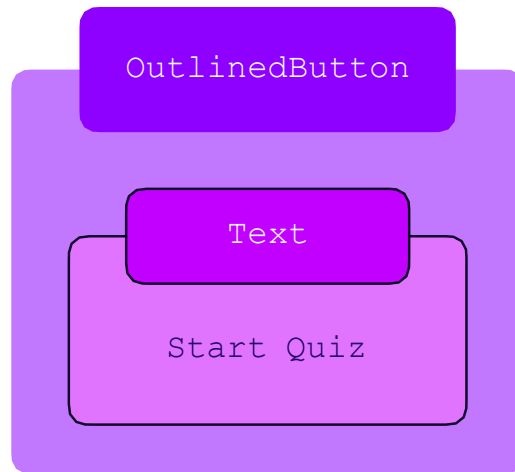
IT'S ALL ABOUT WIDGETS!

**Flutter UIs are created by
combining & nesting
Widgets**



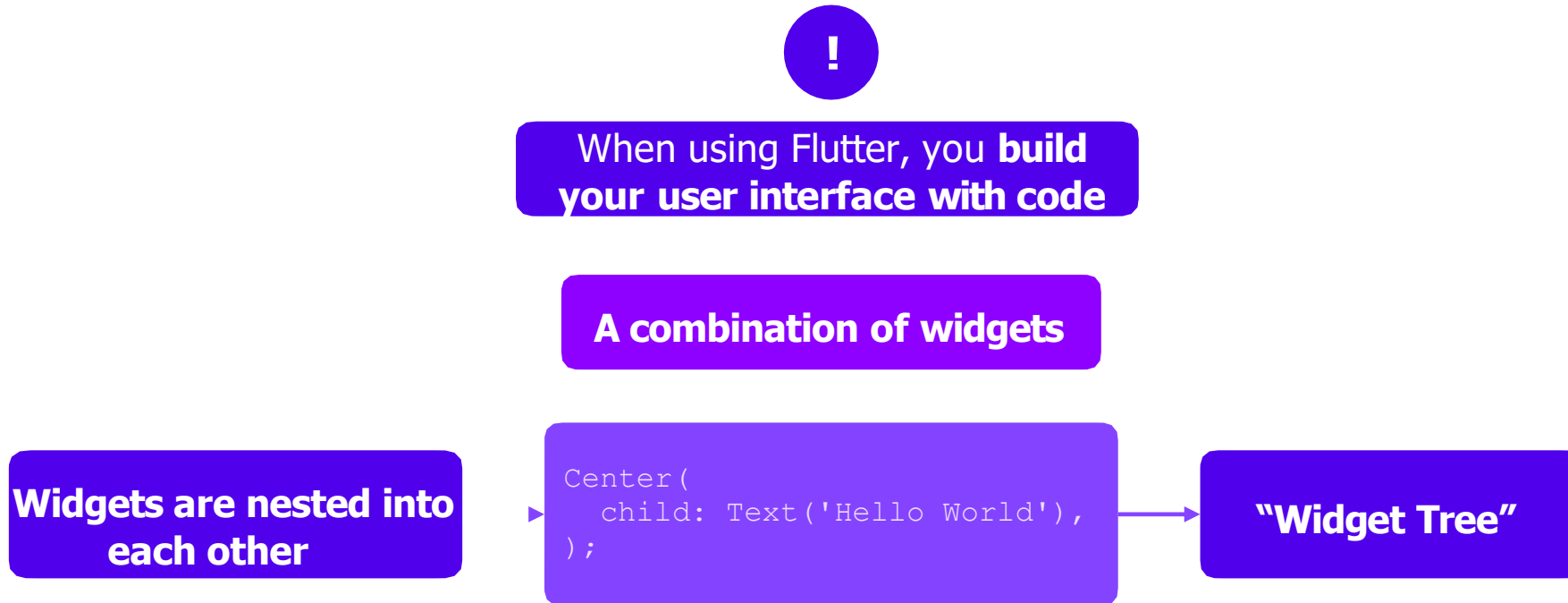
IT'S ALL ABOUT WIDGETS!

Flutter UIs are created by
combining & nesting Widgets



- ✓ **Flutter provides many built-in Widgets**
e.g., Buttons, form inputs, layout widgets, ...
- ✓ **You can also build your own Widgets**
Based on the built-in Widgets

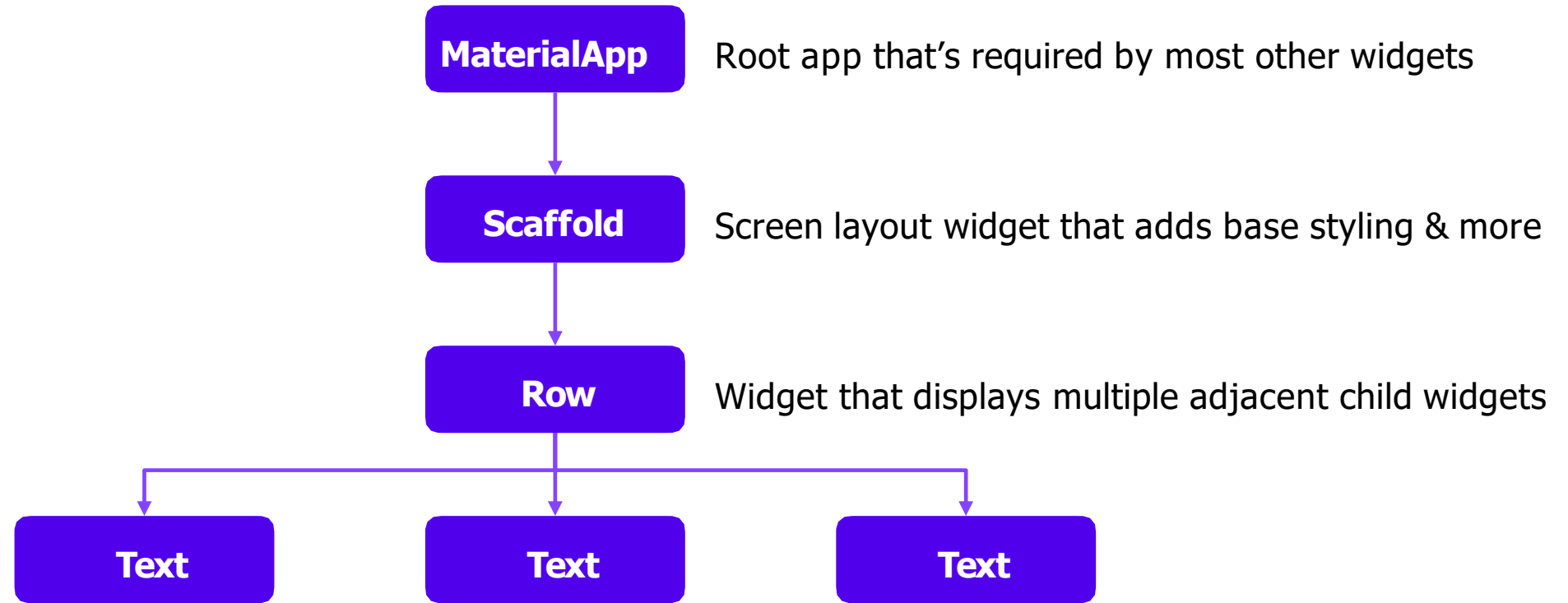
FLUTTER UIS ARE BUILT WITH WIDGETS



Built-in Centerwidget centers its content
horizontally + vertically

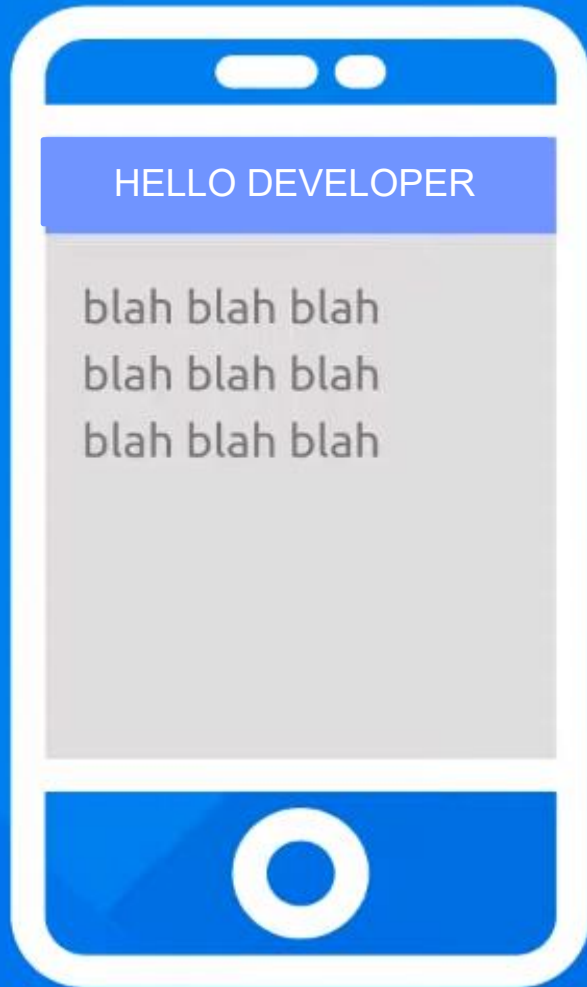
Built-in Textwidget displays some text on the screen

IT'S A WIDGET TREE!

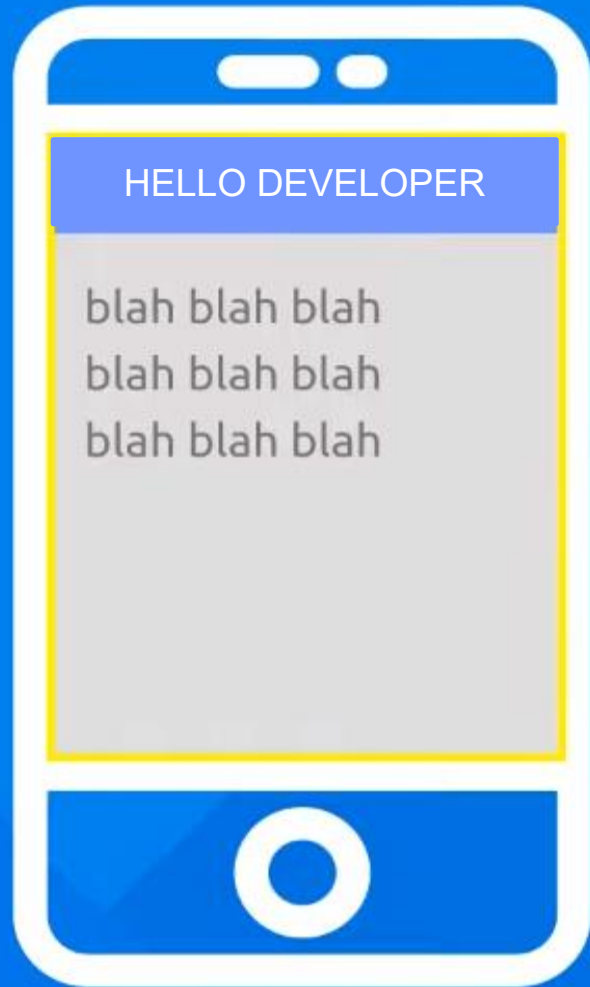


Widgets that display some text on the screen

IT'S A WIDGET TREE!

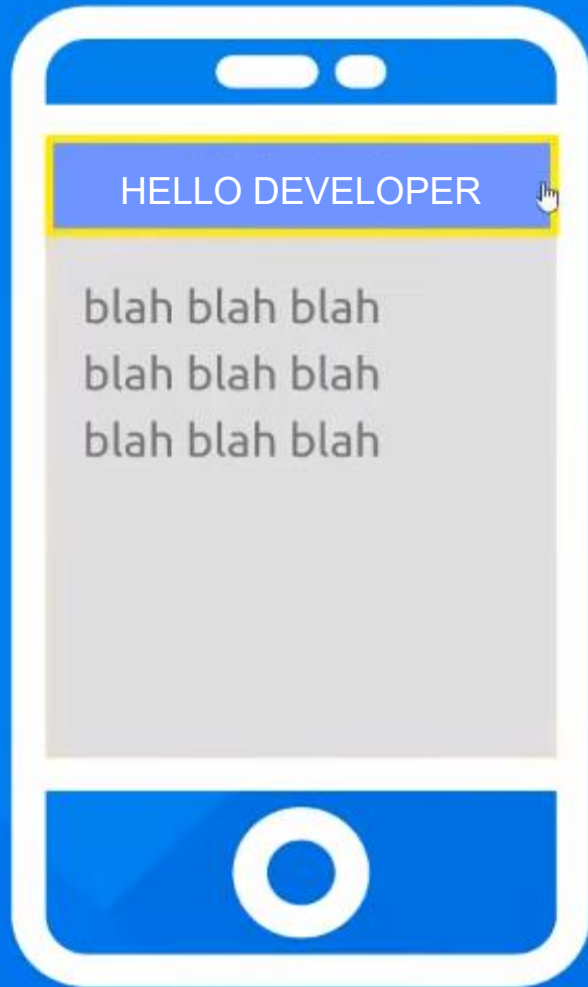


IT'S A WIDGET TREE!



Root Widget

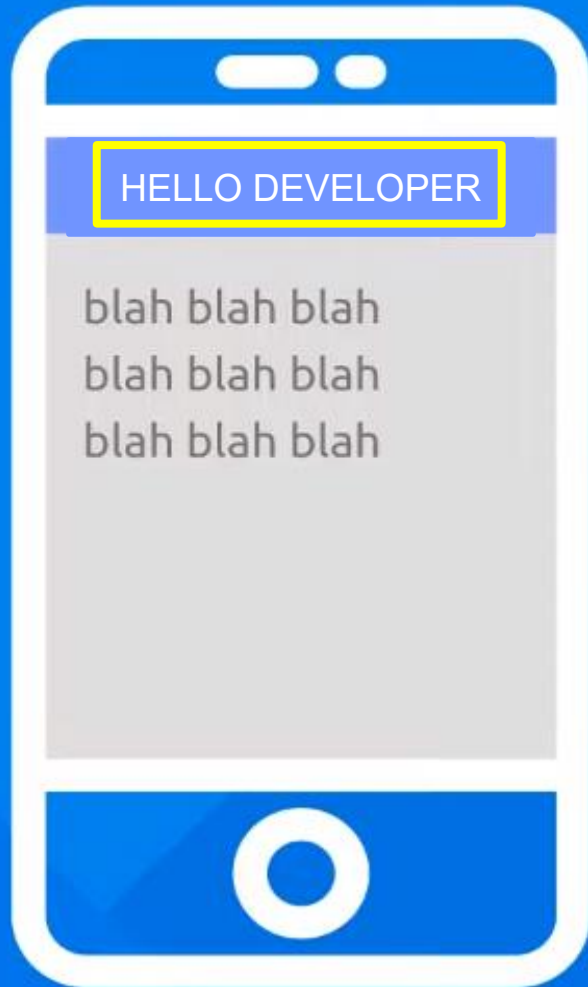
IT'S A WIDGET TREE!



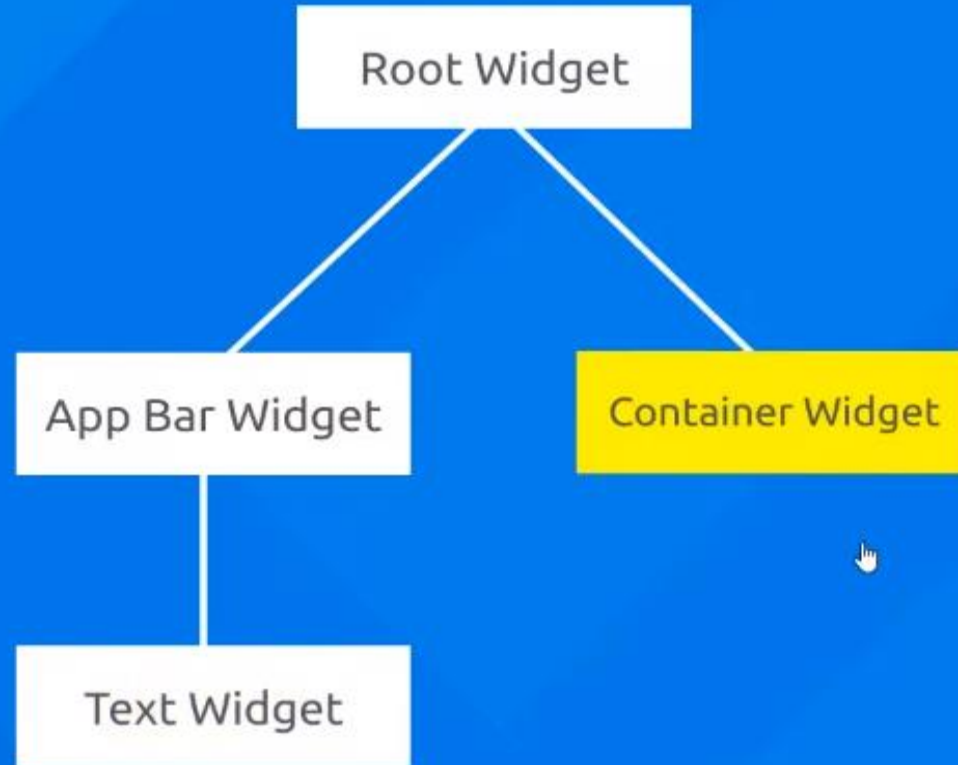
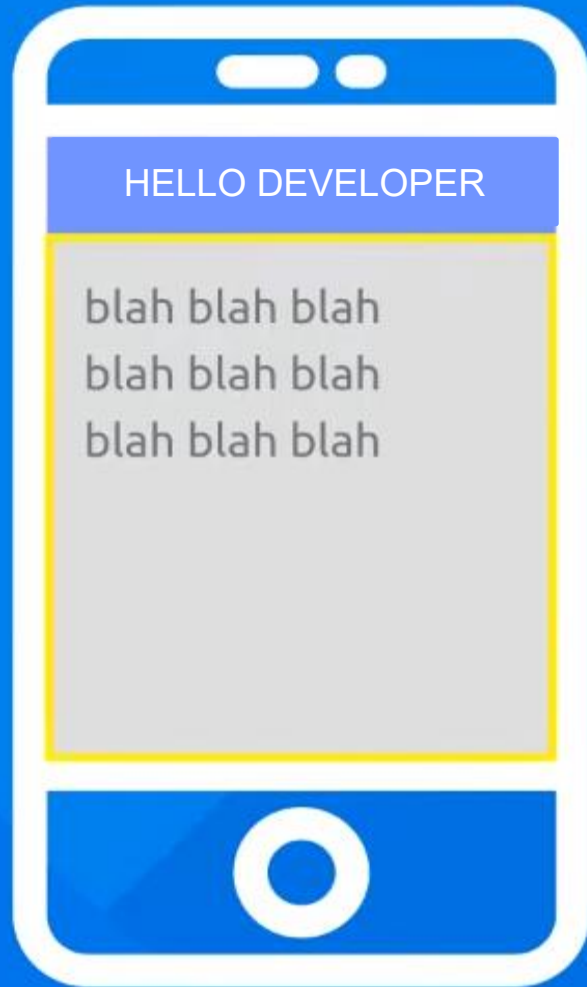
Root Widget

App Bar Widget

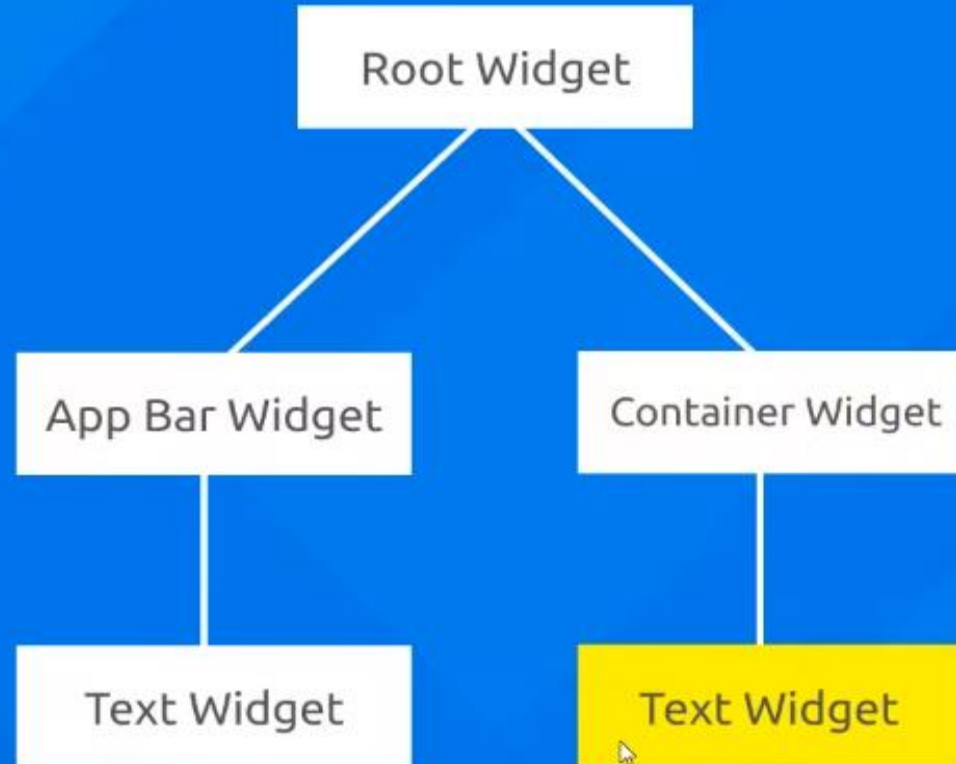
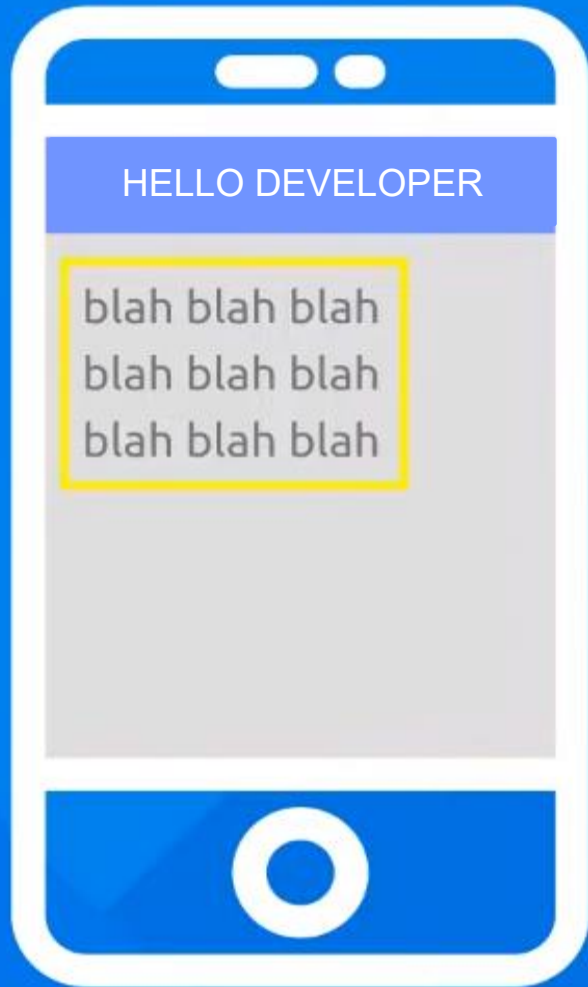
IT'S A WIDGET TREE!



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IT'S A WIDGET TREE!

Text Widget

style
textAlign
overflow
maxLines
... etc

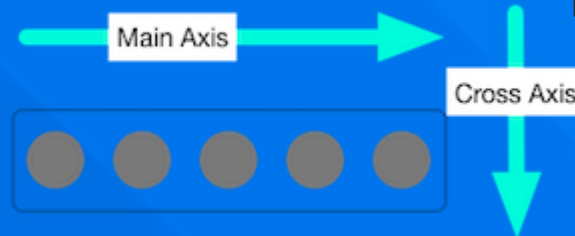
Button Widget

color
elevation
disabledColor
enabled
... etc

Row Widget



Row



Column Widget



Column

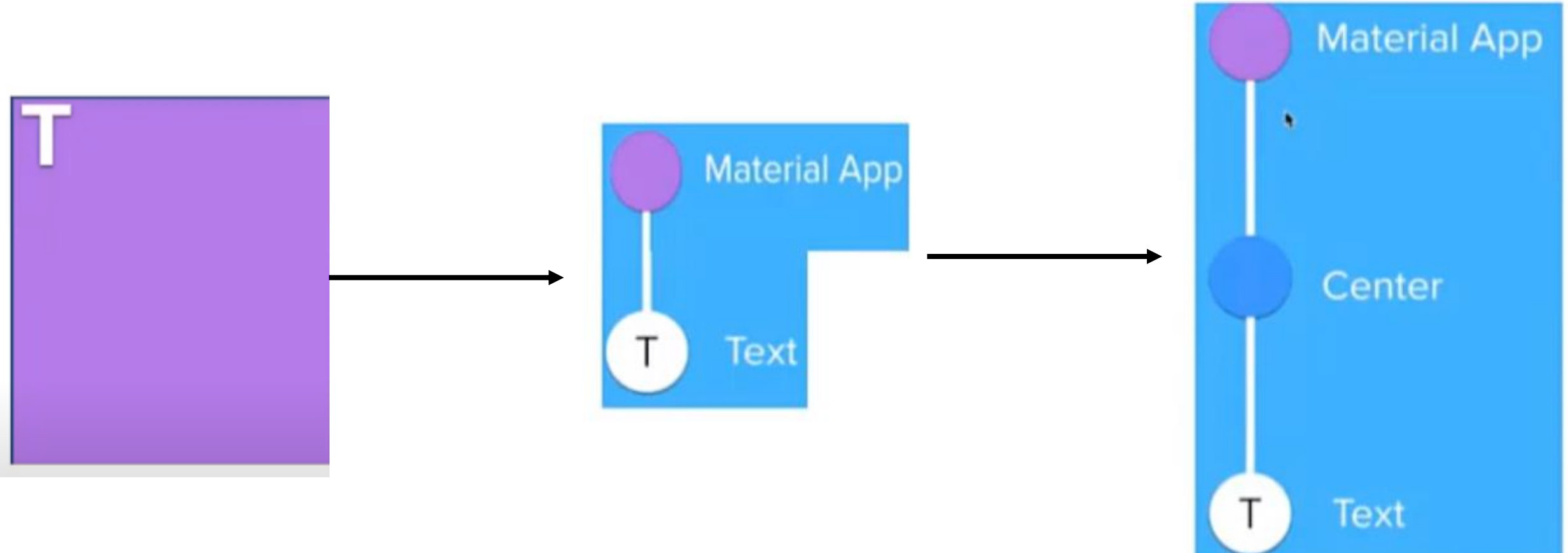


Image Widget

EXAPLES OF ROW AND COLUMN WIDGET



MATERIAL DESIGN APP FROM SCRATCH



Thank You 😊